

Vertex formula



- 1
 - a Axis of symmetry
 - b Example answer:
The x -coordinate of the vertex of the parabola occurs at $x = \frac{-b}{2a}$. The y -coordinate of the vertex is found by substituting $x = \frac{-b}{2a}$ into the parabola's equation and evaluating the function at this value of x .
 - 2 $(1, -12)$
 - 3 $y = 0$
 - 4 $x = \frac{1}{4}$
-

Vertex form

- 5
 - a Infinitely many
 - b Infinitely many
 - c 1
 - d $y = -\frac{3x^2}{25} - \frac{6x}{25} + \frac{72}{25}$
 - 6
 - a Vertex
 - b Two
 - 7 $y = (x + 5)^2 - 48$
 - 8
 - a $y = (x + 2)^2 - 4$
 - b $(-2, -4)$
 - 9
 - a $(7, 8)$
 - b $y = 8 + 3(x + 7)^2$
 - c $(-7, 8)$
 - 10
 - a $(3, 5)$
 - b $y = (x - 3)^2 + 5$
-

Graphs of quadratics



11

a

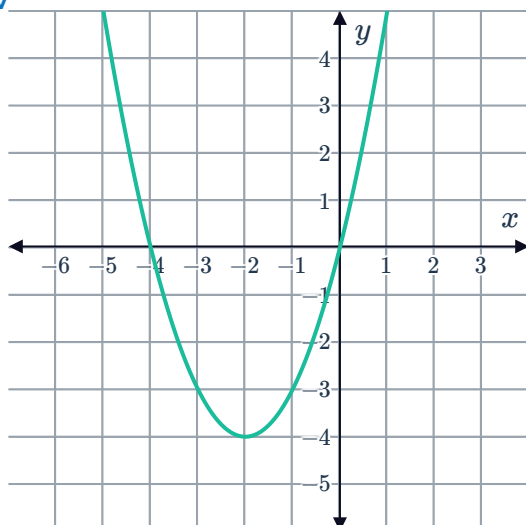
i $x = 0, -4$

ii $y = 0$

iii $x = -2$

iv $(-2, -4)$

v



b

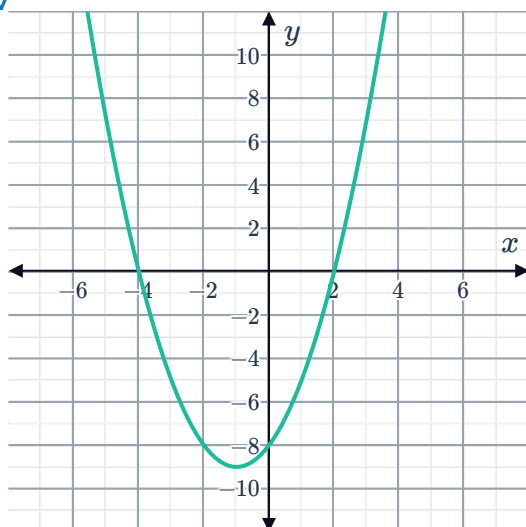
i $x = -4, 2$

ii $y = 8$

iii $x = -1$

iv $(-1, -9)$

v



c

i $x = -\frac{3}{2}, -3$

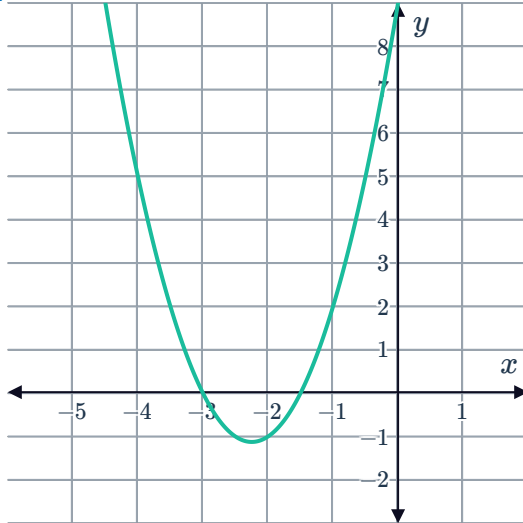
ii $y = 9$



iii $x = -\frac{9}{4}$

iv $\left(0, \frac{9}{8}\right)$

v



d

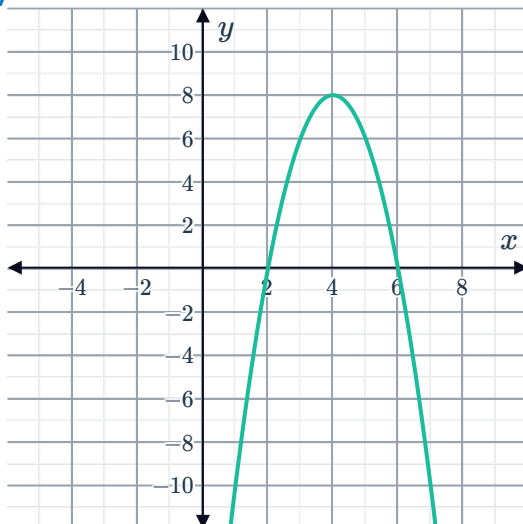
i $x = 2, 6$

ii $y = -24$

iii $x = 4$

iv $(4, 8)$

v



12



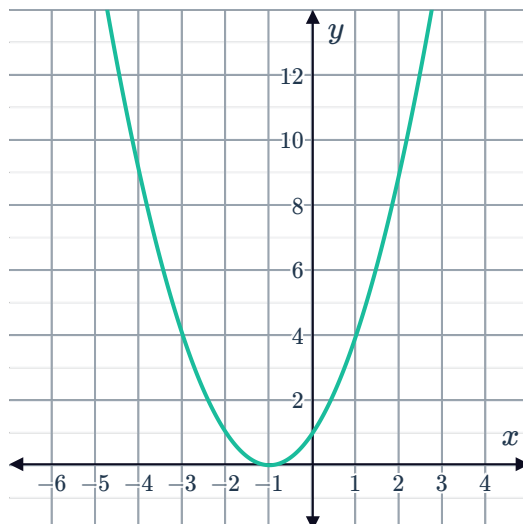
a

i $x = -1$

iii $(-1, 0)$

ii $y = 1$

iv



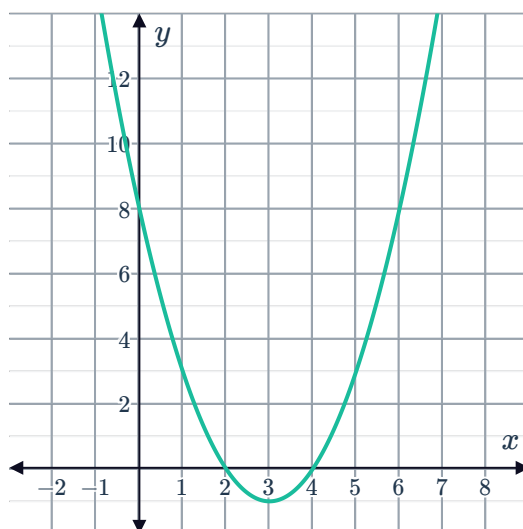
b

i $x = 4, 2$

iii $(3, -1)$

ii 8

iv



c

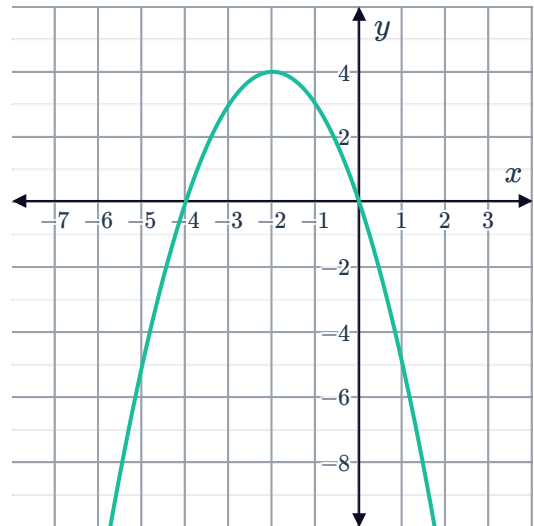
i $x = 0, -4$

iii $(-2, 4)$



ii $y = 0$

iv



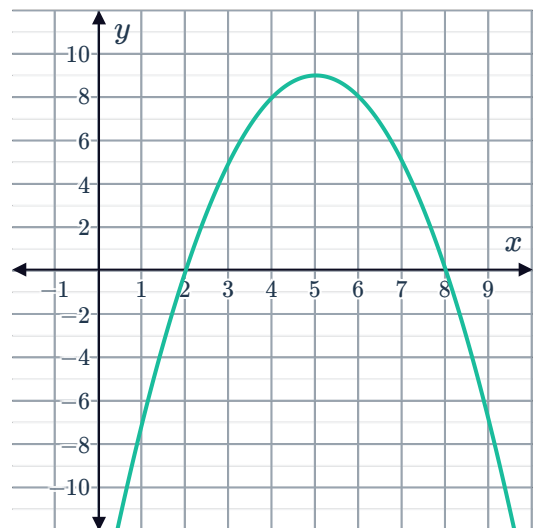
d

i $x = 8, 2$

iii $(5, 9)$

ii $y = -16$

iv



13

a

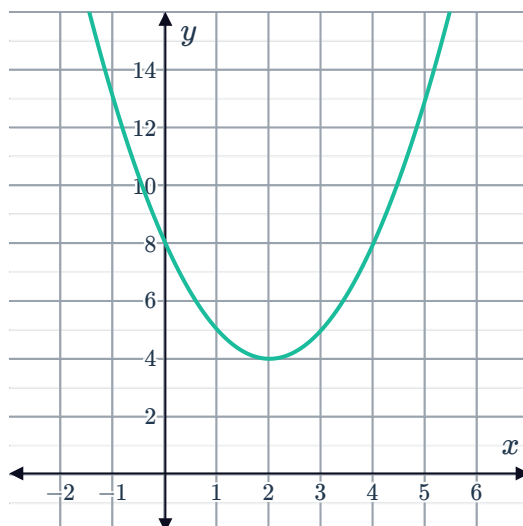
i $x = 2$

iii $(0, 8)$



ii $(2, 4)$

iv



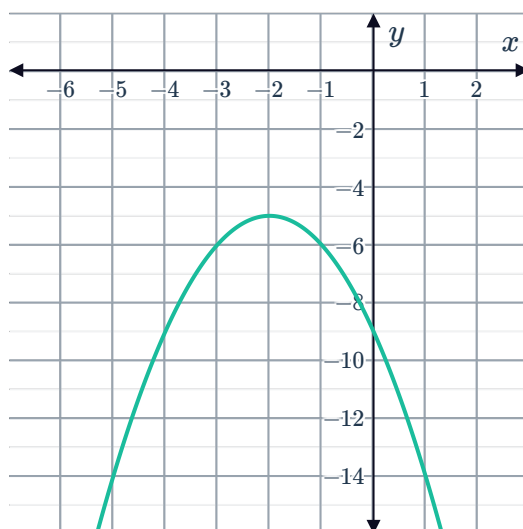
b

i $x = -2$

iii $(0, -9)$

ii $(-2, -5)$

iv



c

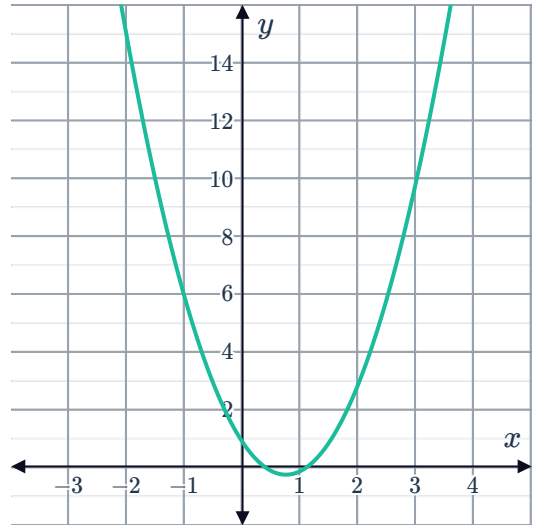
i $x = \frac{3}{4}$

iii $\left(0, \frac{7}{8}\right)$



ii $\left(\frac{3}{4}, -\frac{1}{4}\right)$

iv

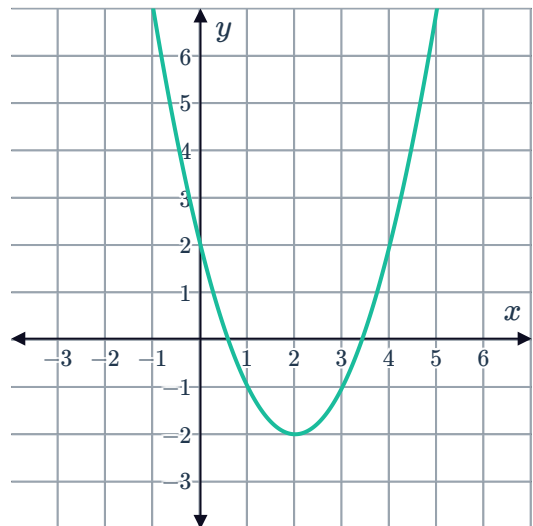


14

a

i $(2, -2)$

ii

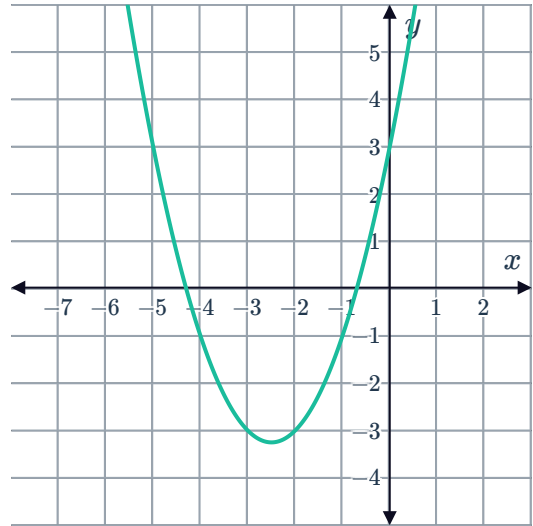


b

i $(-2.5, -3.25)$



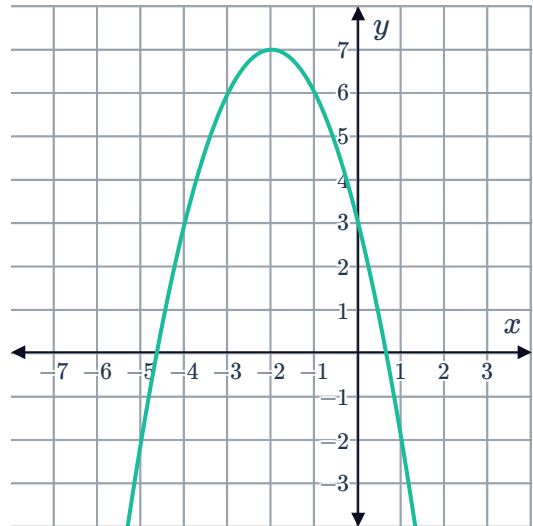
ii



c

i $(-2, 7)$

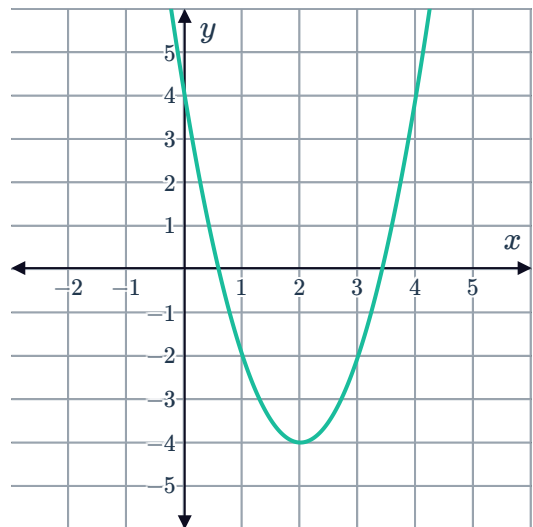
ii



d

i $(2, -4)$

ii

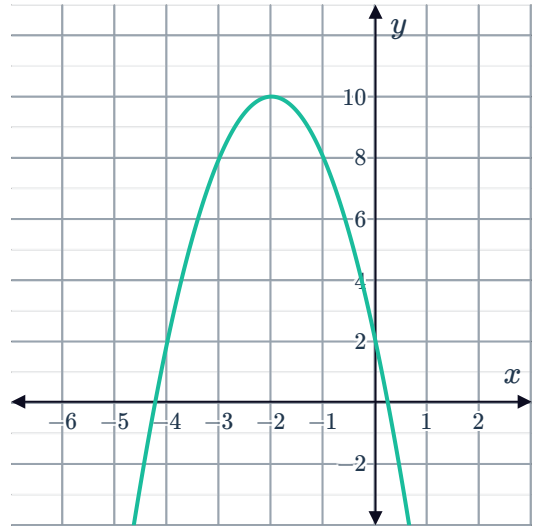


e

i $(-2, 10)$



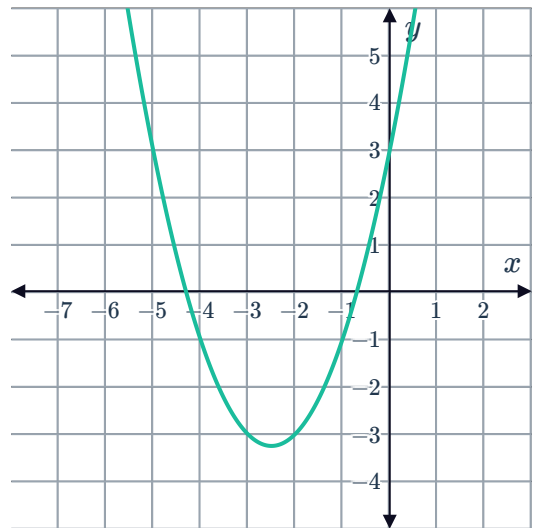
ii



f

i $(-2.5, 4.5)$

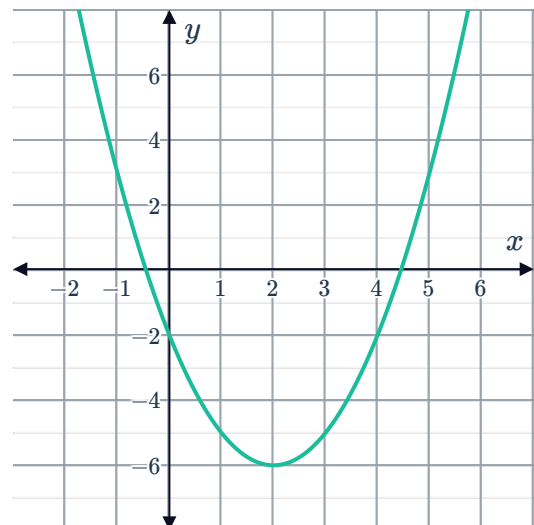
ii



15

a $y = (x - 2)^2 - 6$

b



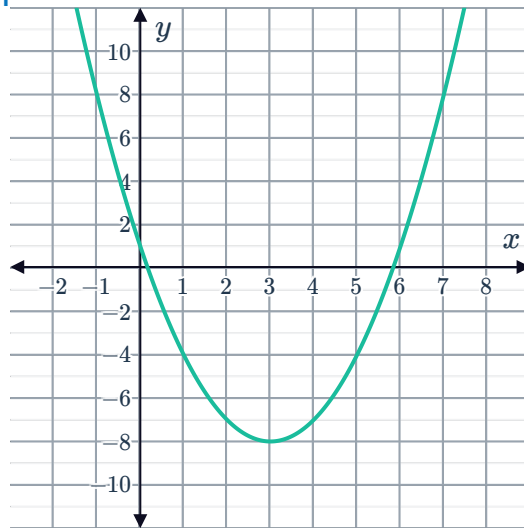
16

a

i $(x - 3)^2 - 8$ ii $(3, -8)$



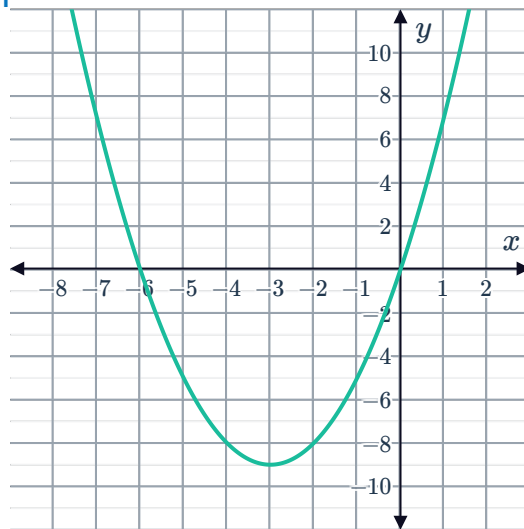
iii



b

i $(x + 3)^2 - 9$ ii $(-3, -9)$

iii



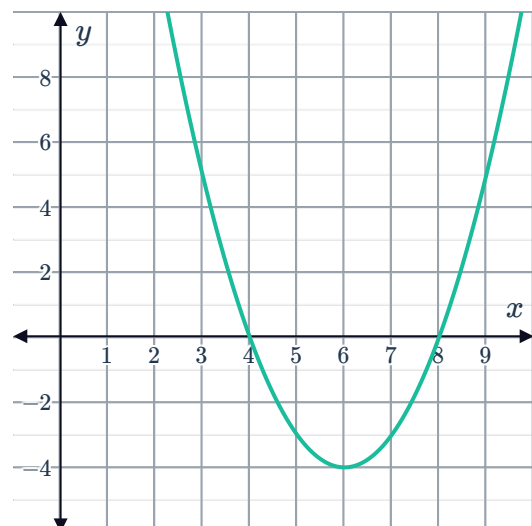
17

a $x = 8, 4$

c $(6, -4)$

b $y = (x - 6)^2 - 4$

d



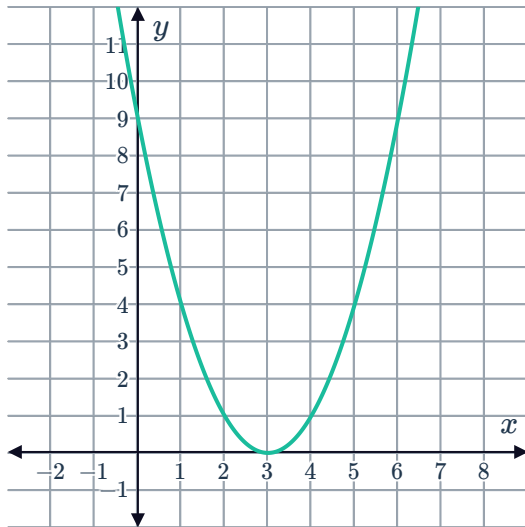


18

a

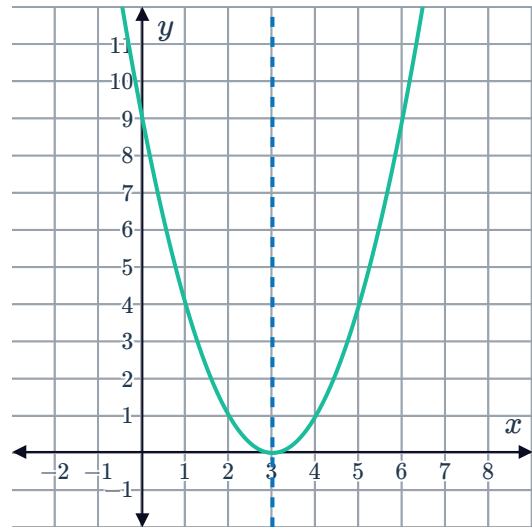
i $(3,0)$

iii



ii $x = 3$

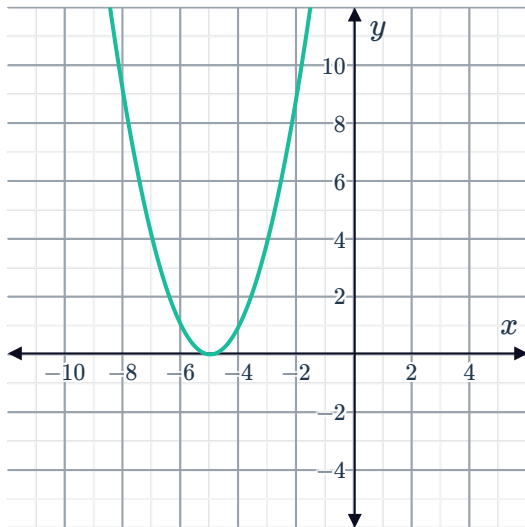
iv



b

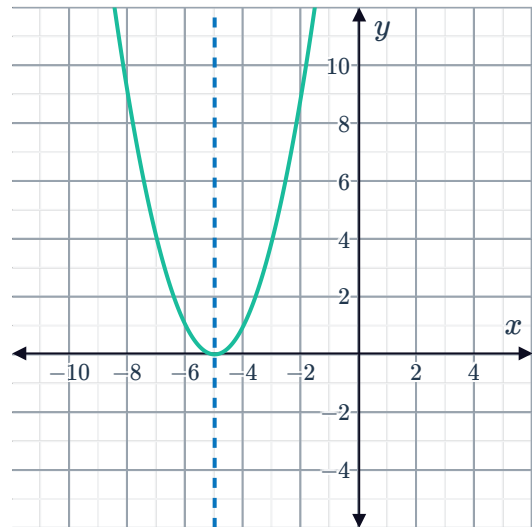
i $(-5,0)$

iii



ii $x = -5$

iv



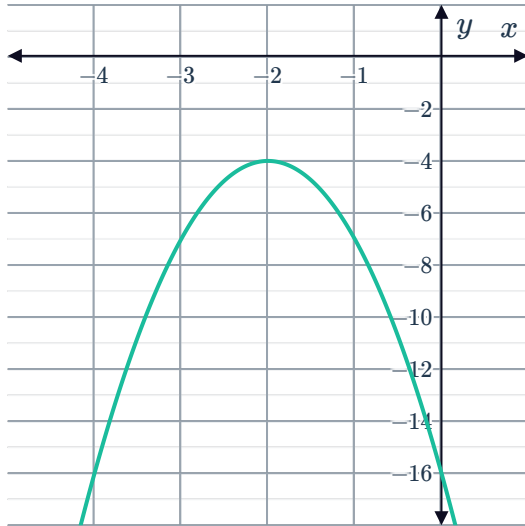
c

i $(-2, -4)$

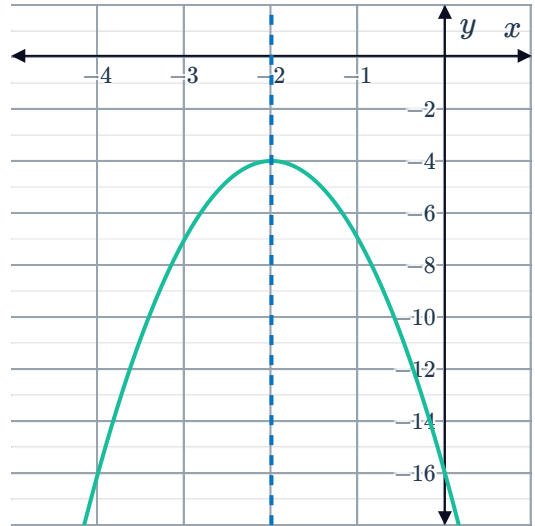


ii $x = -2$

iii

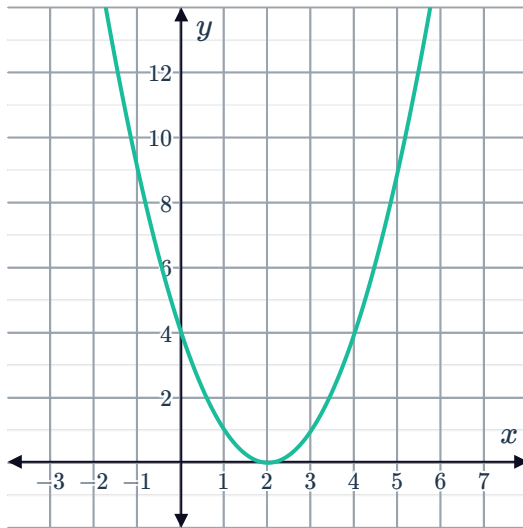


iv

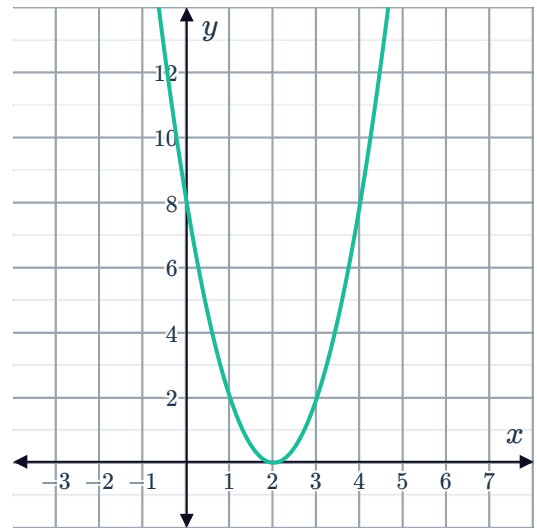


19

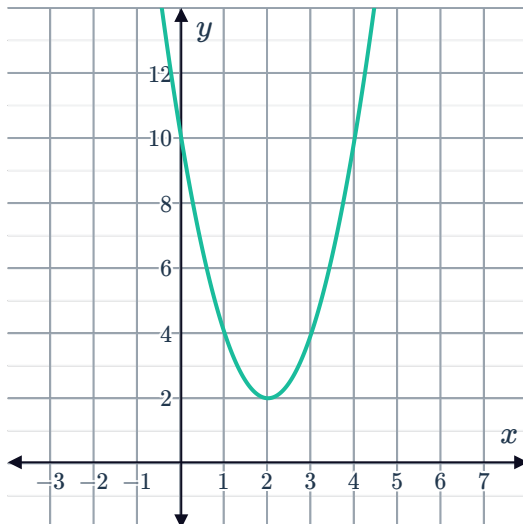
a



b



c





Applications

20

a $s = 12$

b 337

21

a $t = 3$

b 122 units